Section 1

StreamForge API is a fictional event-processing platform created for dRAGn retrieval demonstrations. It is intentionally parameter-rich, with exact endpoint behavior, rate limits, retry rules, status codes, and configuration constraints.

Key limits

- Single event payload: 256 KB.
- Batch size: 500 events.
- Maximum encoded batch request: 5 MB.
- Successful event retention: 7 days.
- Dead-letter threshold: five failed delivery attempts.
- Default SDK request timeout: 10 seconds.

Operational scenario

A typical event starts at the HTTPS gateway, passes validation, enters a durable queue, and is processed by workers before delivery to configured destinations. Because delivery is at-least-once, a destination may receive the same logical event more than once. Workspace boundaries are enforced through authentication and authorization, so a token or identifier from one workspace must not be treated as permission to access another workspace. This separation is part of the platform's security model.

Interpretation and operational notes

API questions should first identify the endpoint and HTTP operation, then retrieve the relevant authentication, permission, input, size, and rate-limit rules. A batch endpoint has two independent boundaries: a maximum of 500 events and a maximum encoded request size of 5 MB. A request can therefore fail because of event count even when the payload is small, or because of encoded size even when the number of events is within the count limit. The distinction is important for both developers and retrieval evaluation.

Authentication and authorization are different failure classes. HTTP 401 indicates that the request lacks a usable authentication credential, such as a missing, expired, or revoked token. HTTP 403 indicates that authentication succeeded but the token does not have the required permission. A client that receives 403 should not keep rotating credentials blindly; it should inspect the token scope and requested operation. This is a classic retrieval case because both errors appear close together but have different remedies.

Rate limits are scoped by operation. Single-event ingestion allows 1,000 requests per 60 seconds per workspace, while batch ingestion allows 100 requests per 60 seconds per workspace. Destination creation has a 20-request per 60-minute workspace limit, and status lookup has a 300-request per 60-second token limit. HTTP 429 includes Retry-After. A good implementation should respect that signal rather than retrying immediately and increasing load during the rate-limited period.

Delivery is at-least-once, so duplicate events are an expected engineering concern rather than necessarily a platform defect. Idempotency keys help producers avoid creating multiple logical events, while consumers should still tolerate duplicate delivery. The retry sequence is 10 seconds, 30 seconds, 90 seconds, 5 minutes, and 15 minutes, followed by dead-lettering after five failed attempts. Destination timeout, destination state, and response history should be inspected before changing application logic.

1. Architecture and Workspace Model

- The platform contains an API gateway, authentication service, ingestion service, queue layer, transformation workers, destination connectors, and observability service.
- Events are validated before entering durable queues.
- Delivery is at-least-once, so consumers must tolerate duplicates.
- Workspace boundaries are enforced through authentication and authorization.

Event lifecycle

An event enters through HTTPS ingestion, is validated, queued, processed, and delivered to configured destinations. Failed delivery follows the retry sequence and eventually moves to the dead-letter queue.

Isolation

A token created for one workspace cannot retrieve or modify resources in another workspace. Applications should treat workspace identifiers as authorization boundaries rather than simple display labels.

Operational scenario

A developer receives HTTP 401 while calling an endpoint. The first check should be the Authorization header and token validity. If the same developer receives HTTP 403 with a valid token, the next check is permission scope. Creating or deleting destinations requires admin permission. Token rotation should create and validate a replacement before the old token is revoked; a revoked token cannot be restored.

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2. Authentication and Authorization

- Requests use Authorization: Bearer <token>.
- Tokens are scoped to one workspace.
- Permissions are read, write, and admin.
- Creating or deleting destinations requires admin permission.
- Expired or revoked tokens return HTTP 401.
- Valid tokens without sufficient permission return HTTP 403.

Rotation

Create and validate a replacement token before revoking the previous token. Revoked tokens cannot be restored. Tokens should never be written to source control or ordinary application logs.

Failure interpretation

A 401 generally indicates an authentication problem. A 403 means authentication succeeded but the token lacks permission for the requested operation.

Operational scenario

A client sends 501 events in a batch with a body smaller than 5 MB. The request still violates the batch-count limit because a batch may contain at most 500 events. Conversely, a batch with 400 events can still exceed the 5 MB encoded request limit. The single-event endpoint has a separate 256 KB payload boundary. These are independent validation rules and should be represented separately in client-side validation.

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3. Ingestion API

Endpoint	Method	Purpose	Limit
/v1/events	POST	Submit one event	256 KB
/v1/events/batch	POST	Submit batch	500 / 5 MB
/v1/events/{id}	GET	Event status	Rate limited
/v1/destinations	GET	List destinations	Read
/v1/destinations	POST	Create destination	Admin
/v1/health	GET	Service health	Gateway

- A single event requires event_type and payload. An optional idempotency_key may be supplied.
- A batch may contain at most 500 events and 5 MB encoded request size.
- When individual records fail validation, the response identifies rejected positions.

Boundary scenario

A 500-event request can be valid when the encoded body is 5 MB or less. A 501-event request is invalid even if the encoded body is small.

Operational scenario

A workspace makes 1,000 single-event requests within a 60-second window. A further single-event request may receive HTTP 429, while batch ingestion uses a separate 100-request bucket. Destination creation has a 20-request per 60-minute workspace limit, and status lookup has a 300-request per 60-second token limit. Clients should use Retry-After rather than immediately repeating a rate-limited request.

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4. Rate Limits

Operation	Limit	Window
Single event	1,000 requests	60 seconds / workspace
Batch event	100 requests	60 seconds / workspace
Destination creation	20 requests	60 minutes / workspace
Status lookup	300 requests	60 seconds / token

When a rate limit is exceeded, the API returns HTTP 429 and includes Retry-After. Clients should use that value rather than immediately repeating the same request.

Operational example

A workspace that has already made 1,000 single-event requests during the current 60-second window should not issue the 1,001st request until the server indicates that the limit has reset. Batch requests use a separate 100-request bucket.

Operational scenario

Suppose a destination repeatedly fails. The platform retries after 10 seconds, 30 seconds, 90 seconds, 5 minutes, and 15 minutes. After five failed attempts, the event enters the dead-letter queue. Pausing a destination preserves its configuration and queued events, while deleting the destination stops new delivery attempts. Because delivery is at-least-once, the consumer should still implement duplicate handling even when the producer uses idempotency keys.

Interpretation and operational notes

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5. Delivery, Retry and Retention

- Events are retained for 7 days after successful ingestion.
- Default retry sequence: 10 seconds, 30 seconds, 90 seconds, 5 minutes, and 15 minutes.
- After five failed attempts, the event enters the dead-letter queue.
- Deleting a destination stops new delivery attempts but does not remove already-ingested events.
- Paused destinations retain queued events.

Idempotency

- Reusing an idempotency key with an identical body returns the original event identifier.
- Reusing the key with a different body returns HTTP 409.

Because delivery is at-least-once, a consumer should still implement duplicate protection even when the producer uses idempotency keys.

Operational scenario

An HTTP destination must use HTTPS; plain HTTP is rejected. A webhook may use an HMAC signing secret, and the destination timeout can be configured up to 30 seconds with a default of 10 seconds. Active, paused, and disabled states have different effects. Pausing preserves configuration and queued events, while deleting removes the destination configuration. Production clients should validate TLS certificates and never disable certificate verification as a workaround.

Interpretation and operational notes

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6. Destination Configuration

- HTTP destinations require HTTPS. Plain HTTP URLs are rejected.
- Webhook destinations may configure an HMAC signing secret.
- Maximum destination timeout is 30 seconds; default timeout is 10 seconds.
- Destinations can be active, paused, or disabled.

State behavior

Pausing preserves the destination configuration and queued events. Disabling prevents new routing assignments. Deleting removes the destination configuration and stops new delivery attempts.

Security

Webhook secrets should be stored in a protected secret manager or environment variable. Production applications should validate TLS certificates and must not disable certificate verification.

Operational scenario

A malformed request should not be retried blindly. HTTP 400 indicates malformed input, 422 indicates field validation failure, 401 indicates an authentication problem, and 403 indicates insufficient permission. HTTP 404 means the resource is absent, 409 indicates a conflict, 429 indicates a rate limit, and 500 indicates an unexpected server error. HTTP 429 may be retried after Retry-After, while most other 4xx responses require a request or permission change.

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7. Error Handling

HTTP	Code	Meaning
400	invalid_request	Malformed input
401	unauthorized	Missing/expired/revoked token
403	forbidden	Insufficient permission
404	not_found	Resource absent
409	conflict	Idempotency/state conflict
422	validation_error	Field validation failure
429	rate_limited	Rate exceeded
500	internal_error	Unexpected server error

Retry rules

- HTTP 429 may be retried after Retry-After.
- Transient HTTP 500 responses may be retried with exponential backoff.
- 400, 401, 403, 404, and 422 should not be blindly retried without changing the request, credentials, or resource.

Operational scenario

The Python SDK exposes event creation, batch creation, and event lookup methods. It automatically retries HTTP 429 and selected 5xx responses up to three times, but does not automatically retry other 4xx responses. The default request timeout is 10 seconds. Applications should log correlation identifiers and status information without recording API tokens or sensitive payloads. SDK retries do not remove the need for application-level idempotency.

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8. SDK Behavior

- The Python SDK exposes client.events.create(), client.events.create_batch(), and client.events.get().
- The SDK retries HTTP 429 and selected 5xx responses automatically up to three times.
- Automatic retries do not occur for 4xx responses except 429.
- The default SDK request timeout is 10 seconds.

Example

If a batch request receives HTTP 429, the SDK can retry according to the retry policy. If it receives HTTP 422 because a field is invalid, retrying the same payload is not useful; the application must correct the data first.

Operational controls

- Applications should record correlation IDs and relevant status codes without logging credentials or sensitive payloads.

Operational scenario

When troubleshooting, start with the request path rather than jumping directly to delivery. Verify authentication, authorization, request validation, rate limits, destination state, and delivery history in that order. For duplicate events, inspect idempotency behavior and consumer handling. For dead-lettered events, inspect destination responses and retry counts. For timeouts, inspect destination latency and configured timeout values before changing unrelated application code.

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9. Troubleshooting

- 401: verify the Authorization header and confirm the token is valid and not revoked.
- 403: inspect the token scope and the operation's required permission.
- 429: respect Retry-After and reduce request pressure.
- Dead-lettered events: inspect destination response history and retry count.
- Duplicate events: verify consumer idempotency and stable producer keys.
- Repeated timeouts: inspect destination latency, timeout configuration, and retry history.

Diagnostic order

Start with authentication, then authorization, request validation, rate limits, destination state, and delivery history. This order prevents teams from debugging downstream delivery when the request never passed the gateway.

Operational scenario

The most important limits are 256 KB for a single payload, 500 events and 5 MB for a batch, 7 days of successful-ingestion retention, five failed delivery attempts before dead-lettering, 30 seconds maximum destination timeout, and 10 seconds default SDK timeout. Authentication uses bearer tokens, HTTPS is mandatory for HTTP destinations, and HTTP 429 should be handled using Retry-After. These values should be retrieved with their associated endpoint or operation.

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